

ESO208 COMPUTATIONAL METHODS FOR ENGINEERS

ASSIGNMENT 2

MAXIMUM MARKS: 30

DUE DATE: 22 AUGUST 2026

Q.1. [15 marks] The saturation concentration of dissolved oxygen in freshwater can be calculated using the following equation:

$$\ln(O_{sf}) = -139.34411 + \frac{1.575701 \times 10^5}{T_a} - \frac{6.642308 \times 10^7}{T_a^2} + \frac{1.243800 \times 10^{10}}{T_a^3} - \frac{8.621949 \times 10^{11}}{T_a^4}$$

where O_{sf} is the saturation concentration of dissolved oxygen in freshwater (mg/L), absolute temperature of water $T_a = T + 273.15$, T is the temperature in °C. The equation is valid for a range $0^\circ\text{C} \leq T \leq 40^\circ\text{C}$.

- (a) Write a code for finding the root of an equation using the bisection method. [10]
- (b) Use your code to find the freshwater temperature at which the saturation concentration of dissolved oxygen is 10 mg/L. [5]

Q.2. [15 marks] The discharge $Q(\frac{m^3}{s})$ in a rectangular channel with width $B(m)$, flow depth $y(m)$, bed slope $S(-)$ and roughness n is given by

$$Q = \frac{\sqrt{S}(By)^{5/3}}{n(B + 2y)^{2/3}}$$

- (a) Write a code for finding the root of an equation using the Newton-Raphson method. [10]
- (b) Calculate the flow depth for which $Q = 5m^3/s$ in a canal with $B = 20\text{ m}$, $S = 0.0002$ and $n = 0.03$. [5]